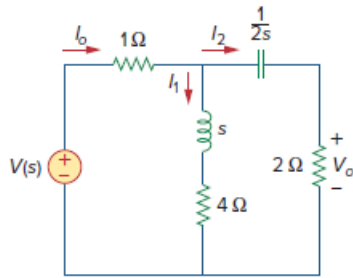


Problem

Find the transfer function $H(s) = I_1(s) / I_o(s)$ in the circuit of Fig. 16.18.

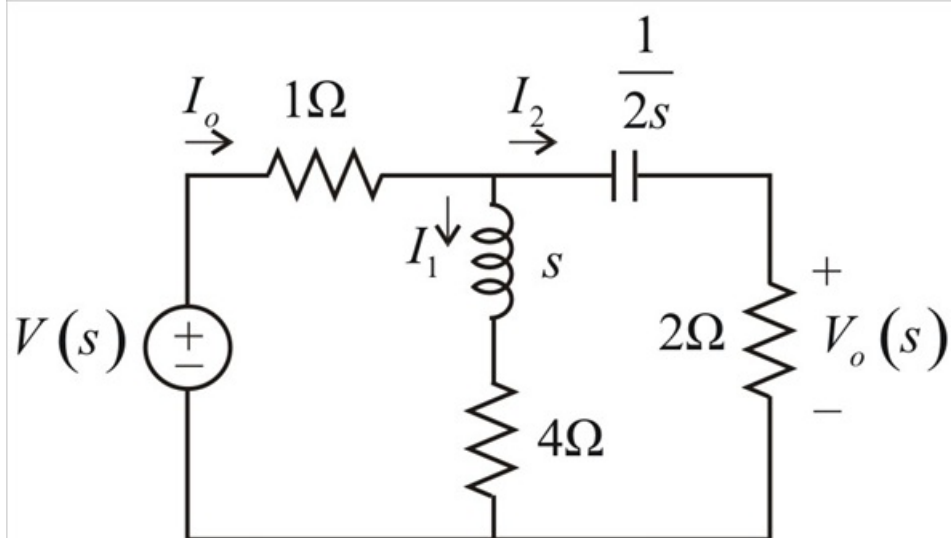
Figure 16.18:



Step-by-step solution

Step 1 of 2

Given circuit



Step 2 of 2

Using current division,

$$I_1 = \left(\frac{2 + \frac{1}{2s}}{2 + \frac{1}{2s} + s + 4} \right) I_o$$

$$I_1 = \left(\frac{4s + 1}{4s + 1 + 2s^2 + 8s} \right) I_o$$

$$\therefore \boxed{\frac{I_1(s)}{I_o(s)} = \frac{4s + 1}{2s^2 + 12s + 1}}$$

Thus, the transfer function $H(s)$ is $\frac{4s + 1}{2s^2 + 12s + 1}$